

Student Portfolios 2026

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The Portfolio

Document individual achievements as a student portfolio, which has the following form:

- PDF
- Format A4 landscape
- 15-25 pages, excluding cover and references
- Max upload size is 20MB
 - Compress larger PDFs, e.g. with https://www.ilovepdf.com/compress_pdf

The PDF Must Contain

- Title page **With name**
- Your selected data visualizations
 - **In the order specified** under “visualization tasks”
 - You decide what to include.
 - One topic may correspond to more or less than 1 page.
 - Contents and weighting of subjects are up to you, as long as you fulfill all requirements.
 - Everything you submit will be graded. Additional work can improve or reduce the grade.
- A reference page (last page), containing
 - Sources for all data sets
 - Sources for third party plots (i.e. the “good” and “bad” examples)
 - All software used ...
 - * AI tools
 - * Python or R packages
 - * Web services (e.g. flourish, datawrapper, etc.)
- **You may use any software/tool.** You may improve the visualizations that have been created in Python or other programs using any additional software (Adobe Illustrator, Inkscape).
- Beyond annotations of your visualizations, text is only necessary if required in a task
- You do not need to submit any program code
- **Submission** and **deadline** see Canvas.

Grading Metric

- Appropriateness of chosen visualization type
- Conformity to the rules of data visualization **Not over clutter, double encoding etc.**
- Statistical and economic correctness
- Detailed quality, i.e. iterations
- Relevance / story
- Consistency of the entire portfolio (topic, design)
- Difficulty and breadth of the portfolio
- Quality and originality of resources

These criteria are presented in decreasing order of importance. Originality of sources cannot, for example, compensate for choosing the wrong type of graph.

Work and Tools

- Work individually.
- Use of Generative AI is expected **to create graphs**.
- Generative AI is not permitted for writing the short texts.

Visualization Tasks

The following tasks should, as a minimum, be fulfilled in your portfolio. Present at least 1 visualization for each task **in the order below**. You may add visualizations at the end. Do not exceed the page limit.

1. Find a bad and/or manipulative visualization. Write a 100-150 word critique of it
2. Create an improved version of the “bad/manipulative” visualization
3. Find a particularly good visualization and write a 100-150 word critique of it
4. Create a visualization about climate change
5. Create a black-and-white visualization (no grey levels)
6. Create a visualization that uses color as an important aesthetic
7. Create a visualization that rigorously maximizes Tufte’s “data-ink ratio” (look it up).
8. Create a visualization that is none of the following: map, bar chart, scatter plot, pie chart, doughnut chart, line chart, box plot, density plot, histogram, radar chart
9. Create a visualization of textual data
10. Create one visualization by hand. Choose plain paper or graph paper and create your viz accordingly. Scan your viz using university photocopiers or make a really good photo of it.
11. Create one data map
12. Create one interactive visualization

- The interactive visualization must be **deployed** at a **public URL**. It must be usable in a standard web browser without downloading or installing anything and without creating an account.
 - In the portfolio, document your interactive visualization using screenshot(s) and mention the public URL.
 - Rigorously apply the definition of “interactive visualization” as given in class. Simple hover or zoom features do not qualify as “interactive”.
13. Document the creation process of one visualization from idea and (hand) sketch to several versions and the final version (add up to 150 words explaining the process if you want)
 14. Optional: Additional visualizations created by you
 15. Describe (using as many words as you require) your favorite classical (i.e. non-AI) tools for data visualization. These can be (amongst others): Python/R packages, other software programs, web sites, cheat sheets, books.
 16. Describe (using as many words as you require) your favorite AI tools for data visualization. These can be (amongst others): models, web sites, software that uses or orchestrates AI
 17. Describe (using as many words as you require) your favorite resources of inspiration for data visualizations. These can be (amongst others): data sources, news organizations with good/bad data visualizations, blogs, other sources of inspiration.
 18. Reference page: sources for all data used, sources for “good” and “bad” visualizations, all tools used, disclosure of use of generative AI for each individual graph

Notes

- For number 1 and 3, do not use any visualization discussed in class.
- Number 12 has to be interactive, all others can be static or interactive at your choice.